POORNACHANDRAN A

AI & Machine Learning | RAG Systems | Scalable Backend and AI Applications

6385322821 poorna2006poorna@gmail.com GitHub LinkedIn Tiruchirappalli, India

PROFESSIONAL SUMMARY

Computer and Communication Engineering student at Amrita Vishwa Vidyapeetham specializing in AI systems, large language model applications, and scalable backend architectures Interested in research at the intersection of LLM systems, information retrieval, and intelligent AI infrastructure.

PROJECTS

Neural Search Engine (Retrieval-Augmented Generation System) | Github Sep 2025 - Oct 2025

AI Systems Developer

- Designed and implemented a production-grade **Retrieval-Augmented Generation (RAG)** system enabling **semantic search**, **multi-document reasoning**, and **LLM-based answer synthesis** over large document collections
- Built a scalable retrieval pipeline using **Qdrant vector database** with **semantic embeddings**, chunk indexing, and score-based filtering.
- Engineered a **Map-Reduce style multi-document reasoning architecture** to generate and synthesize per-document answers using an **LLM** for accurate cross-document insights
- Developed an automated **Python ingestion pipeline** for document parsing, **chunking with overlap**, **embedding generation**, and vector indexing.
- Implemented a **Next.js conversational interface** with multi-chat sessions and persistent query history for interactive document exploration.

Technologies / Tools Used : *python, LLM APIs, Vector Embeddings, Qdrant, REST APIs, Next.js, TypeScript, Node.js, Express*

AI-Driven 5G Network Digital Twin for Predictive Network Optimization | Github Feb 2026 - Present

System Architect & AI Developer

• Designed and implemented a **Digital Twin architecture** to simulate cellular network behavior and enable **predictive analysis** without affecting real infrastructure.

• Developed a **telemetry simulation** generating time-varying metrics such as **user density**, traffic demand, latency, and resource utilization to model realistic network dynamics.

• Built a centralized **Network State Model** that aggregates telemetry from multiple base stations to maintain a **real-time network representation**.

• Implemented a deterministic **simulation engine** for **what-if analysis** and **AI-assisted network optimization** to forecast congestion scenarios and recommend **load-balancing strategies**.

• Exposed the system through a **FastAPI-based service layer** with modular architecture, **configuration management**, and **structured logging** for scalable experimentation.

Technologies / Tools Used : *Python, FastAPI, Uvicorn, Simulation Modeling, Object-Oriented System Design, REST APIs, Structured Logging, Configuration Management*

EDUCATION

Amrita Vishwa Vidyapeetham Jul 2024 - Present
B.Tech CCE *CGPA : 8.49/10*
Pursuing a Bachelor of Technology in Computer and Communication Engineering with a focus on Artificial Intelligence, Machine Learning,scalable AI systems. and Web Development
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Artificial Intelligence, Probability and Statistics.

WORK EXPERIENCE

Web Development Intern | Internship Dec 2025 - Jan 2026
ShadowFox *Remote*
• Developed a fully responsive portfolio web application using HTML5, CSS3, JavaScript, React.js, and Tailwind CSS, implementing a modular component-based architecture for scalable UI development.
• Applied modern frontend engineering practices including responsive design, UI optimization, and cross-browser compatibility testing to ensure consistent performance across devices.
• Deployed the application on Vercel and maintained version control using Git, ensuring stable deployment, build optimization, and production-ready hosting.
Technologies / Skills Used **:** *HTML5, CSS3, JavaScript, React.js, Tailwind CSS, Git, Vercel*

SKILLS

Programming Languages :	C, JavaScript, Python, TypeScript
Frameworks & Libraries :	FastAPI, React.js, Next.js, Tailwind CSS, Vue, Express, Flask, REST APIs, Node.js
Tools & Platforms :	Docker, Git, GitHub, Postman, Uvicorn, Vercel, VS Code
Databases :	Qdrant, MongoDB, PostgreSQL, MySQL, SQLite
Soft Skills :	Teamwork, Problem Solving, Adaptability, Critical Thinking
Languages :	English

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification Dec 2025
DeepLearning.AI and Stanford University (Coursera)
Certification Url : https://coursera.org/verify/BCU03JO4WY1F

Introduction to Software Engineering Dec 2025
IBM (Coursera)
Certification Url : https://coursera.org/verify/L8PWAD9LO8W1